

# GEORGE LARSON

Email: george.g.larson@pm.me · Phone: 412-945-0101 · Location: Pittsburgh, PA

Director of Technology with 25+ years building and operating software, infrastructure, and manufacturing systems. I have led teams at Sony Electronics, scaled a medical billing startup to national reach, programmed PLCs on production lines, and directed technology for a mid-size manufacturer. I do Linux platforms, DevOps automation, ecommerce, and security programs. I stay close to the keyboard even when the job is strategic, and I use AI tooling where it actually helps.

## KEY METRICS

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- 36 public assets scanned continuously via HostedScan + OWASP ZAP with 33 findings closed in the past 60 days and zero critical/high issues in backlog.
- 36 active GitLab projects (11 users, 2 groups), 2,204 merge requests, 739 CI pipelines (avg 565 successes / 58 fails) and 16 managed SSH keys.
- Mattermost workspace with 7 teams, 40 channels, ~1.0M lifetime posts, 313 calls, and alert channels wired to uptime, upgrade, helpdesk, GitLab, and web-critical events.
- Zammad helpdesk processed 1,487 tickets YTD (~150–206/month in Jun–Aug) with ≥97% close rate, 68% resolved inside 24 hours (56% inside 8 hours), zero reopen events.
- Uptime Kuma monitors 31 services (GitLab, Helpdesk, Shopify APIs, FTP, chat, docs, etc.) as part of 24×7 observability feeding Mattermost.

## CURRENT FOCUS

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- Seeking a full-time technology leadership role where I can build, ship, and lead teams hands-on.
- Building systems tools in Rust and Go (netgrep, termshark) for network analysis and security.
- Operating a multi-agent architecture (nullclaw + ironclaw) with tiered inference, agent-to-agent routing, and security-hardened perimeter design.
- Strengthening security programs, compliance boundaries, and knowledge management systems that keep data and people protected.

## CORE STRENGTHS

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### Software Development and Engineering

- Full stack delivery across Python, Go, Rust, C#, TypeScript, PHP, Perl, C++, SQL.
- Code review, automated testing, and performance tuning for legacy and greenfield systems.

### DevOps and Infrastructure Automation

- CI and CD design, infrastructure as code, container orchestration, and repeatable releases on self-hosted runners.
- Observability, incident response, high-availability design (replication, backups, CDN), and pragmatic SLO tracking for hybrid environments.

### Technology Leadership and Strategy

- Strategic roadmaps that connect executive goals to team-level delivery.
- Calm stewardship for blended internal, offshore, and vendor teams.

### Security and Privacy

- Network hardening, IAM, HostedScan/ZAP/Nikto assessment loops, and zero-trust rollouts.
- Regulated-data guardrails and tabletop response drills.

### Data Engineering and Automation

- OCR digitization, ETL pipelines, semantic search, and knowledge-base design.
- Workflow automation using scripting, open models, and RAG pipelines where they deliver measurable ROI.

## OPEN SOURCE PROJECTS

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### nullclaw Agent Infrastructure (Zig), Deployment and Integration (2026 to Present)

- Deployed nullclaw as a public-facing AI concierge ("Nully") on a hardened perimeter box with Haiku for conversational triage and Sonnet for code analysis, demonstrating tiered inference at the edge.

- Built the full communication stack: Ergo IRC server, self-hosted gamja web client, WebSocket proxied through Cloudflare, Let's Encrypt TLS with automated renewal.
- Designed agent-to-agent routing between nullclaw (public doorman) and ironclaw (private personal agent) across separate systems with distinct security boundaries.
- The agent answers visitor questions by examining public GitHub repositories and preloaded context rather than rephrasing static resume text.

#### **netgrep (Rust), Creator and Maintainer (2026 to Present)**

- Modern replacement for ngrep: live packet capture and pcap analysis with TCP stream reassembly, TLS 1.2/1.3 decryption, HTTP/2 and DNS parsing, and an interactive terminal UI.
- Maintains ngrep-compatible CLI flags; outputs colored text, JSON, hex dump, or pcap.

#### **termshark (Go), Contributor and Maintainer (2026 to Present)**

- Modernized fork of termshark, a terminal UI for Wireshark/tshark.
- Refactored architecture, improved test coverage, updated to Go 1.22+ idioms, and added an experimental browser-based web interface.

## **WORK EXPERIENCE**

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#### **Director of Technology, FM Expressions (2018 to Present)**

- Led technology strategy for a mid-size manufacturing company serving ecommerce channels.
- Improved database performance by 93 percent while driving critical systems to 99 percent uptime; cut MySQL slow-query volume by 96 percent to accelerate incident response.
- Built a knowledge management platform (Wiki.js, Gitea, Qdrant, ETL pipelines) that captures production logs, transcripts, and operator notes so teams can search, surface, and act on institutional knowledge in real time.
- Resolved conflicting manufacturing documentation, created a style guide and glossary, and instituted monthly and quarterly review cycles to keep safety and operations guidance consistent.
- Reduced a 1.03 GB legacy software repository (4,656 binary artifacts) to a clean, LFS-ready codebase, eliminating clone failures and enabling sustainable development practices.
- Hardened hybrid infrastructure with Authentik SSO, HashiCorp Vault, automated TLS renewal, firewall hardening (iptables, UFW, CrowdSec, fail2ban), DMARC/SPF/DKIM enforcement, HostedScan/OWASP ZAP sweeps across 36 public assets, and Red Sift OnDMARC (p=reject, 1.5M emails/month at 99.48% compliance) to keep critical/high findings at zero and close 33 issues in the past 60 days.
- Run GitLab (36 projects, 11 users, 739 CI pipelines) and Mattermost (7 teams, 40 channels, ~1M posts) as the collaboration core tied into Redmine (22 cross-functional projects), Zammad (~200 tickets/month with 68% closed inside 24 hours), and Uptime Kuma (31 monitors feeding alert channels).
- Maintained high-availability platforms: dual MySQL replicas, VMware ESXi with Datto backups, Active Directory, Cisco Meraki switching, SonicWall/pfSense firewalls, and a staged CDN migration from self-hosted edges to Contabo and Cloudflare.
- Built the Mashy internal dashboard plus tooling such as Psono vault, PrivateBin, phpMyAdmin/Adminer, DBEaver, knowledge mining utilities, and status pages so teams access credentials, data stores, and observability in one click.
- Driving adoption metrics for the knowledge stack to quantify time-to-answer gains for production and customer operations.
- Captured Meteor print-head PCAPs and tuned Nikto, Metasploit, and custom fuzz rigs to secure industrial protocols while migrating remote access from VPN to zero-trust policies.
- Implemented Agile delivery, ticketing, and analytics to align product, operations, and finance on throughput, defect, and margin targets.

#### **Software Developer (2017)**

- Departed after company practices conflicted with professional engineering standards.

#### **Senior Software Engineer and Manager, Universal SmartComp (2009 to 2016)**

- Helped scale the company from a 50 person startup to a national leader in medical billing.
- Designed EDI frameworks, ACH processing, ERP integrations, and a medical payer portal.
- Led internal and offshore teams across development and Linux infrastructure.
- Acted as security response lead and Linux SME; migrated project management to Redmine with custom plugins.

- Built OCR bill entry validation that reduced errors and sped turnaround.

### **Software Engineer and Team Manager, Sony Electronics (2004 to 2009)**

- Directed software development, logistics, and technical operations for the Pittsburgh Customer Satisfaction Center.
- Designed and ran the TiVo firmware upgrade station for the United States market.
- Built hardware testing suites for CD drives, set-top boxes, PCBs, televisions, cameras, and MP3 players.
- Created an AS/400 parts ordering interface that saved millions in operational spend.
- Managed more than 50 combined technical and non-technical staff.
- Developed a VT-52 terminal emulation system used for inventory management.

### **Maintenance and Automation Technician, T. Marzetti Company (2000 to 2004)**

- Led second shift maintenance for robotics, conveyors, and packaging machinery.
- Programmed PLCs and tuned sensors, actuators, and industrial controls on the production line.

### **Earlier Career Highlights (1990s to 2000s)**

- Software developer and webmaster for non-profit organizations.
- Engineer for The Sounding Board, a psychotherapeutic BBS in Columbus, Ohio; built custom C++ modules for Major BBS and kept one of the first internet-connected BBS systems alive.
- Contract developer for Daifuku USA working on material handling support systems.
- Two-time Business Professionals of America chapter president; regional competitor in extemporaneous speech.

## **VOLUNTEER WORK**

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- Systems administrator for a volunteer Linux community, digital radio station, and open shell environment.
- Long-term support for animal shelters, including events, outreach, and facilities work.
- OCR and digitization project contributor for historical legal documents.

## **TECHNICAL SKILLS**

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**Languages:** Python, Go, Rust, Zig, C#, TypeScript, PHP, Perl, C++, Bash, SQL

**Web and Data:** MySQL, PostgreSQL, WordPress, Redmine, Gitea, Wiki.js

**Infrastructure:** Linux, Apache, Nginx, Caddy, IIS, Kubernetes, VMware ESXi, Datto backups, Cisco Meraki, SonicWall, pfSense, CrowdSec, Let's Encrypt/ACME, Cloudflare CDN

**Cloud:** AWS, Azure, Google Cloud, Proxmox

**Security and Identity:** Authentik, HashiCorp Vault, Psono, CrowdSec, HostedScan, OWASP ZAP, Nikto, Metasploit, SSL/TLS, iptables, UFW, fail2ban, DMARC/SPF/DKIM, DefectDojo, OpenVAS, zero-trust network design

**Collaboration:** Agile, Scrum, Kanban, Redmine, Zammad, GitLab, Mattermost, CryptPad

**Mail Systems:** iRedMail, Postfix, Dovecot

**Commerce:** Shopify, custom ecommerce platforms, payment integrations

**Data and Automation:** RAG design, OCR pipelines, semantic search, ETL, knowledge base design

**AI Agent Infrastructure:** nullclaw, ironclaw, multi-agent routing, tiered inference, model selection (Haiku/Sonnet/Opus), LLM cost optimization, agent security boundaries

**Systems and Networking:** TCP/IP analysis, packet capture, PLC programming, industrial controls, IRC server administration

**Monitoring & Dashboards:** Uptime Kuma, HostedScan, Red Sift OnDMARC, Dashy internal portal

## **EDUCATION AND PROFESSIONAL DEVELOPMENT**

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- Self-taught from C++ books as a teenager; built BBS modules in production before attending college.
- Continuous learner across hundreds of online courses, lectures, and certifications (see LinkedIn for details).
- Currently studying: Stanford CS336 — Language Modeling from Scratch (2025).
- Additional coursework spanning systems engineering, network security, database internals, and cloud architecture through platforms including Coursera, MIT OpenCourseWare, and vendor certification programs.